

What is a PRD

In product management and development, **PRD** stands for **Product Requirements Document**. It is a comprehensive guide that outlines the purpose, features, functionality, and behavior of a new product or a significant feature.

The PRD serves as a "single source of truth" that aligns cross-functional teams (product managers, designers, developers, and stakeholders) on what needs to be built and why.

Key Components of a PRD

A PRD typically includes:

- **Overview and Objectives:** A high-level summary of the product's vision, goals, and how it aligns with overall business strategy.
- **Target Audience & Problem Statement:** Identification of the customer personas and the specific pain points or problems the product aims to solve for them.
- **Features and Functionality:** A detailed list of what the product must do, often broken down into user stories and acceptance criteria.
- **User Experience (UX) Requirements:** Descriptions or visuals (wireframes, mockups) of the intended user flow and interface design.
- **Technical Specifications:** Details for the engineering team, such as the technology stack, performance expectations, security, and integrations.
- **Release Plan & Timeline:** Key milestones, dependencies, and a target release date.
- **Metrics for Success:** Key Performance Indicators (KPIs) used to measure the product's performance and impact after launch. It should be possible to measure quantitatively that using the product contributes to value as defined by the audience. To this end, the product must produce logs of activity that can be analyzed to compute automatic metrics that measure indication of value.
- **Assumptions and Constraints:** Potential risks, limitations (e.g., budget, resources, regulatory), and assumptions made during planning.

The PRD is a living document that is updated throughout the development lifecycle to reflect new insights and feedback, helping manage expectations and prevent *scope creep*.

To make the PRD a living document, we will keep it as a component of our GitHub repository and update it through a set of reviewed PRs (Pull Requests).

PRDs and LLMs

Large Language Models (LLMs) can significantly streamline and enhance the process of working on Product Requirements Documents (PRDs) by acting as powerful assistants for generation, refinement, analysis, and validation.

Here is how LLMs can help with writing, improving, and critiquing PRDs:

To Write PRDs (Generation & Drafting)

LLMs excel at generating structured content quickly, drastically reducing the time required to create a first draft:

- **Generating a Structured Template:** An LLM can instantly generate a standard PRD template with relevant sections (e.g., Objectives, Target Audience, Features, Metrics) tailored to a specific industry or product type.
- **Drafting Section Content:** By providing a brief prompt or a few bullet points, an LLM can flesh out full sections. For example, a prompt like "Draft a problem statement for a new task management app aimed at remote teams" can produce a polished, professional description.
- **Creating User Stories:** LLMs can transform high-level requirements into structured user stories in the standard format: "As a (user persona), I want to (action), so that I can (benefit/value)". This saves significant time in agile planning.
- **Summarizing Inputs:** LLMs can ingest meeting notes, customer feedback transcripts, or market research data and synthesize them into concise summaries suitable for the "Problem Statement" or "Assumptions" sections of the PRD.

To Improve PRDs (Refinement & Enhancement)

LLMs can act as an editor and a writing coach, enhancing the clarity and quality of the document:

- **Enhancing Clarity and Conciseness:** LLMs can rephrase technical jargon into plain language, ensuring the document is easily understood by all stakeholders (engineering, design, marketing).
- **Improving Tone and Consistency:** They can adjust the tone to be formal, encouraging, or objective, and ensure consistent terminology is used throughout the document.
- **Proofreading and Editing:** LLMs catch grammatical errors, typos, and syntax issues faster and sometimes more effectively than standard grammar tools.
- **Formatting and Structure:** They can suggest improvements to the document's structure, ensuring a logical flow from high-level objectives down to specific technical requirements.

To Critique PRDs (Validation & Analysis)

LLMs can simulate different perspectives and identify potential weaknesses or gaps in the document:

- **Identifying Gaps and Ambiguity:** An LLM can be prompted to review a finished draft and ask questions a developer or designer might have, such as, "Does this feature

description clearly define acceptance criteria?" or "What use cases are missing from this section?".

- **Checking for Alignment and Consistency:** LLMs can cross-reference different sections to ensure the features proposed in one section directly address the objectives outlined earlier in the document.
- **Simulating Stakeholder Reviews:** An LLM can adopt a persona, such as an "overburdened engineer" or a "skeptical VP of Marketing," to provide critical feedback on feasibility, marketability, or resource constraints.
- **Validating Metrics:** LLMs can suggest relevant KPIs or evaluate whether the proposed "Metrics for Success" directly measure the achievement of the stated goals.

Large Language Models (LLMs) can act as assistants for product managers, significantly enhancing the process of working on Product Requirements Documents (PRDs) by streamlining drafting, improving content quality, and offering critical feedback.

Workflow

We describe here a specific workflow to prepare and maintain the PRD of the product using LLMs based on using GitHub as a shared repository among the team.

1. We will store the PRD related files under a folder in the Repo named PRD.
2. All the files will be stored in Markdown format (.md) [Why MD as format for LLMs](#)
3. We will start with the following files:
 1. **DraftPRD.md** - the high level ideas, brainstorming level.
 2. **PRD.md** - the *live* PRD for the project - it will be updated over time using Pull Requests.
4. We will then refine the toplevel PRD file into different perspectives:
 1. **UserStories.md** - the user stories, broken down by user persona and features.
 2. **DataModel.md** - the data model description
 3. **Glossary.md** - the project terminology to be used consistently when describing functionality
 4. **AcceptanceCriteria.md** - the list of criteria, KPIs and metrics that will be used to determine the product is ready for release.

Using LLMs to Enhance Requirements

LLMs excel at translating unstructured information into structured, comprehensive documents, reducing the time spent on initial drafting and documentation, helping adopt best-practices and increasing consistency when multiple authors collaborate on the requirements.

- **Generating First Drafts from Notes:** Product managers can input brainstorming notes, meeting summaries, or high-level goals and ask the LLM to structure them into a formal PRD draft using a predefined template (e.g., as user stories with "Given/When/Then")

acceptance criteria).

To this end, we will store our original notes in the repo under the **DraftPRD.md** file.

- We will then use the LLM to review, expand, suggest alternatives for the Draft PRD. This will be used to generate or update the **PRD.md** file.
- **Creating User Stories and Acceptance Criteria:** LLMs can break down high-level features into detailed user stories and suggest corresponding acceptance criteria, ensuring thoroughness and clarity for the development team.
We will use this functionality to initialize the **UserStories.md** and **AcceptanceCriteria.md** files.
- **Ensuring Consistency:** LLMs can enforce consistent terminology, tone, and formatting across all documents by adhering to project-wide style guides, making PRDs easier to read and follow. To this end, it helps to create a specific repository with the project specific terminology which we will call **glossary.md**

LLMs can function as "super editors" to refine and optimize the language and structure of a PRD, leading to clearer communication and better alignment.

- **Simplifying Technical Jargon:** LLMs can rephrase complex technical details into more straightforward language, making the document accessible to non-technical collaborators (e.g., marketing, sales, or executives).
- **Summarization:** They can generate concise summaries of long PRDs or specific sections for different audiences (e.g., an executive summary), ensuring efficient communication of key points.
- **Identifying Gaps and Inconsistencies:** LLMs can scan a PRD for missing information, logical inconsistencies, or conflicting requirements, prompting the product manager to address these potential issues before development begins.
- **Suggesting Success Metrics:** LLMs can propose relevant Key Performance Indicators (KPIs) based on the stated objectives and industry benchmarks, helping to define clear, measurable goals for the new product or feature.

LLMs can simulate different perspectives to provide a multi-faceted review, which acts as a valuable testing method before a human team review:

- **Role-Playing Persona:** A PM can prompt the LLM to "Review this PRD from an engineering perspective and flag any technical concerns," or "What questions would a UX designer ask about this feature?". Other persona can be the technical support team, the marketing team, etc.
- **Brainstorming Edge Cases:** LLMs can help identify potential edge cases or error scenarios that human teams might miss during the initial planning phase, enhancing the robustness of the requirements.

Human in the Loop

While LLMs offer benefits, especially for beginner product managers, human oversight remains crucial. Product managers must use LLMs as assistants, not replacements for critical thinking, user research, or final decision-making. LLMs may "mask unknowns" with generic content or produce overly rigid and detailed documents. This must be actively detected and pruned away.